

PE-EC801C — Error Correcting Codes

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Module 1: Linear Block Codes

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1. Introduction to Block Coding

Error control coding is a technique used to detect and correct transmission errors introduced by noisy communication channels. In **block coding**, a stream of information bits is partitioned into blocks of size k . The channel encoder maps each k -bit message block into a larger n -bit codeword by adding $n - k$ redundant parity-check bits.

Key Coding Definitions

- **Code Word (c):** The output block of the channel encoder consisting of n bits.
- **Block Length (n):** The total number of bits in a codeword.
- **Message Length (k):** The number of actual information bits in the block.
- **Redundancy ($n - k$):** The number of check bits added for error detection/correction.
- **Code Rate (R):** The ratio of information bits to total bits, representing transmission efficiency:

$$R = \frac{k}{n}$$

1.1 Vector Space Structure of Codewords

The set of all binary n -tuples forms a vector space V_n over the binary Galois Field $\text{GF}(2) = \{0, 1\}$. There are 2^n possible vectors in V_n . A binary block code C of length n with 2^k codewords is a subset of V_n . If C is a vector subspace of V_n , it is called a **linear block code**. This algebraic structure implies that the sum (modulo-2 addition) of any two codewords in C is also a valid codeword in C :

$$\forall c_1, c_2 \in C \implies c_1 \oplus c_2 \in C$$

1.2 Weight and Distance Metrics

Error correction capability depends on the distance between codewords. We define the following metrics over the binary field:

Weight and Distance Definitions

- **Hamming Weight ($w(x)$):** The number of non-zero elements in a vector $x \in V_n$.
- **Hamming Distance ($d(x, y)$):** The number of positions in which two vectors x and y differ. This is equivalent to the weight of their modulo-2 sum:

$$d(x, y) = w(x \oplus y)$$

- **Minimum Distance (d_{min}):** The smallest Hamming distance between any two distinct codewords in the code C :

$$d_{min} = \min\{d(c_i, c_j) \mid c_i, c_j \in C, c_i \neq c_j\}$$

For any linear block code, the minimum distance is equal to the minimum weight of any

non-zero codeword in the code:

$$d_{min} = \min\{w(c) \mid c \in C, c \neq 0\}$$

1.3 Error Detection and Correction Capabilities

Error Bounds

Let d_{min} be the minimum distance of a block code C .

1. **Error Detection:** A code can detect up to t_d random errors per codeword if:

$$d_{min} \geq t_d + 1$$

2. **Error Correction:** A code can correct up to t_c random errors per codeword if:

$$d_{min} \geq 2t_c + 1 \implies t_c = \left\lfloor \frac{d_{min} - 1}{2} \right\rfloor$$

2. Systematic Linear Block Codes

A block code is **systematic** if the message bits appear unaltered at the beginning (or end) of the codeword, followed by the parity-check bits. A systematic codeword vector c can be partitioned as:

$$c = [m_0, m_1, \dots, m_{k-1}, p_0, p_1, \dots, p_{n-k-1}] = [m \mid p]$$

2.1 The Generator Matrix G

The encoding process is defined by a $k \times n$ matrix called the **Generator Matrix** G . Every codeword is generated by multiplying the k -bit message vector m by G :

$$c = m \cdot G$$

For a systematic linear block code, G is structured as:

$$G = [I_k \mid P]$$

where I_k is the $k \times k$ identity matrix (generating the message part) and P is a $k \times (n - k)$ parity-check coefficient matrix (generating the redundant part).

$$G = \begin{bmatrix} 1 & 0 & \dots & 0 & p_{0,0} & p_{0,1} & \dots & p_{0,n-k-1} \\ 0 & 1 & \dots & 0 & p_{1,0} & p_{1,1} & \dots & p_{1,n-k-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & p_{k-1,0} & p_{k-1,1} & \dots & p_{k-1,n-k-1} \end{bmatrix}$$

The parity bits are computed by:

$$p_j = \sum_{i=0}^{k-1} m_i p_{i,j} \pmod{2}$$

2.2 The Parity Check Matrix H

The parity check matrix H is an $(n - k) \times n$ matrix used to verify if a received vector is a valid codeword. By definition, any valid codeword $c \in C$ must satisfy the orthogonal relation:

$$c \cdot H^T = \mathbf{0}$$

For a systematic code where $G = [I_k \mid P]$, the parity check matrix is:

$$H = [P^T \mid I_{n-k}]$$

where P^T is the transpose of the parity matrix and I_{n-k} is the $(n - k) \times (n - k)$ identity matrix.

Verification of Orthogonality

Let us verify the relationship $G \cdot H^T = \mathbf{0}$:

$$G \cdot H^T = [I_k \mid P] \begin{bmatrix} P \\ I_{n-k} \end{bmatrix} = I_k \cdot P \oplus P \cdot I_{n-k} = P \oplus P = \mathbf{0} \pmod{2}$$

Since $c = m \cdot G$, we have:

$$c \cdot H^T = (m \cdot G) \cdot H^T = m \cdot (G \cdot H^T) = m \cdot \mathbf{0} = \mathbf{0}$$

This confirms that the parity check matrix successfully screens for codewords.

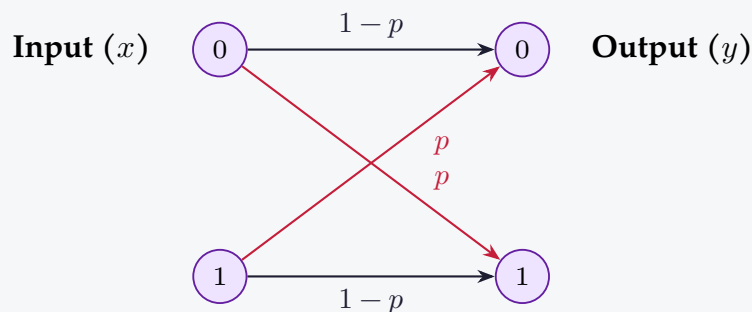
3. Optimum Decoding for the Binary Symmetric Channel

Let c be the transmitted codeword and $r = [r_0, r_1, \dots, r_{n-1}]$ be the received vector. Due to channel noise, $r = c \oplus e$, where $e = [e_0, e_1, \dots, e_{n-1}]$ is the binary error vector ($e_i = 1$ indicates a bit inversion error at position i).

3.1 The Binary Symmetric Channel (BSC)

The BSC is a discrete memoryless channel with binary input, binary output, and a crossover probability p ($p < 0.5$).

Binary Symmetric Channel Model



3.2 Maximum Likelihood (ML) Decoding

An optimum decoder minimizes the probability of decoding error. The maximum likelihood decoding rule selects the codeword $c^{(i)}$ that maximizes the conditional probability $P(r | c^{(i)})$:

$$P(r | c^{(i)}) = p^{d(r, c^{(i)})} (1 - p)^{n - d(r, c^{(i)})}$$

Since $p < 0.5 \implies 1 - p > p$, we can rewrite this by taking the logarithm:

$$\ln P(r | c^{(i)}) = d(r, c^{(i)}) \ln \left(\frac{p}{1 - p} \right) + n \ln(1 - p)$$

Since $\ln(p/(1 - p)) < 0$ for $p < 0.5$, maximizing $P(r | c^{(i)})$ is mathematically equivalent to minimizing the Hamming distance:

ML Decoding Rule: Choose $c \in C$ that minimizes $d(r, c)$

Thus, over a BSC, Maximum Likelihood decoding simplifies to **minimum distance decoding**.

4. Syndrome Decoding on Symmetric Channels

Instead of calculating the distance from r to all 2^k codewords, we can use the parity check matrix to compute a **Syndrome** vector S .

The Syndrome

The syndrome of a received vector r is defined as:

$$S = r \cdot H^T$$

4.1 Properties of the Syndrome

1. $S = \mathbf{0}$ if and only if r is a valid codeword.
2. The syndrome depends **only on the error pattern** e , not on the transmitted codeword c :

$$S = r \cdot H^T = (c \oplus e) \cdot H^T = (c \cdot H^T) \oplus (e \cdot H^T) = \mathbf{0} \oplus (e \cdot H^T) = e \cdot H^T$$

3. Two received vectors r_1 and r_2 have the same syndrome if and only if they differ by a valid codeword.

4.2 The Standard Array and Coset Leaders

A standard array partitions the 2^n possible received vectors into a table of 2^{n-k} rows (called **cosets**) and 2^k columns.

Standard Array Construction Algorithm:

1. Place all 2^k valid codewords in the first row, starting with the all-zero codeword $c_0 = \mathbf{0}$ on the far left.

- From the remaining unused vectors in V_n , choose a vector e_1 of minimum weight. Place it in the first column of the second row. This vector is the **coset leader**.
- Fill the second row by adding the coset leader e_1 to each codeword: $e_1 \oplus c_i$.
- Repeat this process: choose a minimum weight unused vector e_j as the next coset leader, and fill the row with $e_j \oplus c_i$, until all 2^n vectors are placed.

Standard Array Layout

Coset Leader (e_j)	Codeword $c_0 = 0$	Codeword c_1	...	Codeword c_{2^k-1}
$e_0 = 0$	0	c_1	...	c_{2^k-1}
e_1	e_1	$e_1 \oplus c_1$	...	$e_1 \oplus c_{2^k-1}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$e_{2^{n-k}-1}$	$e_{2^{n-k}-1}$	$e_{2^{n-k}-1} \oplus c_1$	...	$e_{2^{n-k}-1} \oplus c_{2^k-1}$

Each row (coset) corresponds to a unique syndrome. The coset leader is the **most likely error pattern** for that syndrome because it has the minimum weight in its coset.

4.3 Syndrome Decoding Algorithm

Syndrome Decoding Steps

Step 1: Compute the syndrome of the received vector: $S = r \cdot H^T$.

Step 2: Find the coset leader e that matches this syndrome: $e \cdot H^T = S$.

Step 3: Correct the received vector to obtain the decoded codeword:

$$\hat{c} = r \oplus e$$

5. Hamming Codes

Hamming codes are a class of single-error-correcting linear block codes. For any integer $m \geq 2$, there exists a Hamming code with the following parameters:

Hamming Code Parameters

- **Parity bits:** $n - k = m$
- **Code length:** $n = 2^m - 1$
- **Message length:** $k = 2^m - 1 - m$
- **Minimum distance:** $d_{min} = 3$
- **Error Correction:** Corrects all single errors ($t_c = 1$).

5.1 Structure of the Parity Check Matrix

The parity check matrix H of a Hamming code has m rows and $2^m - 1$ columns. The columns of H consist of all possible non-zero binary m -tuples, arranged in any order. In a systematic Hamming code:

$$H = [P^T \mid I_m]$$

where the columns of P^T consist of all non-zero binary m -tuples with weight ≥ 2 .

5.2 Decoding of Hamming Codes

When a single error occurs at position i , the received vector is $r = c \oplus e_i$, where e_i has a 1 at index i and zeros elsewhere. The syndrome is:

$$S = r \cdot H^T = (c \oplus e_i) \cdot H^T = e_i \cdot H^T = h_i^T$$

where h_i^T is the transpose of the i -th column of H . Therefore, the syndrome S is **identical to the column of H corresponding to the error location**. If the columns of H are arranged in order of their decimal equivalents, S directly represents the index of the corrupted bit in binary.

6. Weight Enumerators and McWilliams Identities

The performance of a block code is determined by its **Weight Distribution**. Let A_i be the number of codewords in C with Hamming weight i . The set $\{A_0, A_1, \dots, A_n\}$ is the weight spectrum of the code.

6.1 The Weight Enumerator Polynomial

The weight distribution can be expressed as a polynomial:

Weight Enumerator Polynomial

$$A(z) = \sum_{i=0}^n A_i z^i = A_0 + A_1 z + A_2 z^2 + \dots + A_n z^n$$

where $A_0 = 1$ (the all-zero codeword).

6.2 The McWilliams Identities

The McWilliams identities relate the weight enumerator polynomial $A(z)$ of a linear block code C of length n to the weight enumerator polynomial $A^\perp(z)$ of its **dual code $C^\perp$** . The dual code $C^\perp$ is the set of all vectors orthogonal to every codeword in C , having dimension $n - k$ and generator matrix H .

McWilliams Identity Theorem

For a binary $[n, k]$ linear code C with weight enumerator $A(z)$ and dual code $C^\perp$ with weight enumerator $A^\perp(z)$:

$$A^\perp(z) = \frac{1}{2^k} (1+z)^n A\left(\frac{1-z}{1+z}\right)$$

Alternatively, writing as a homogeneous polynomial in two variables x and y :

$$W_{C^\perp}(x, y) = \frac{1}{2^k} W_C(x + y, x - y)$$

This identity is useful because it allows us to find the weight distribution of a code with large k by analyzing its dual code which has a smaller dimension $n - k$ (e.g., finding the weight distribution of a (127, 120) Hamming code by analyzing its (127, 7) dual code).

7. Perfect Codes

For any block code correcting up to t errors, the sphere of radius t centered around each codeword must be disjoint. The total volume of these spheres cannot exceed the size of the vector space 2^n .

7.1 The Hamming Bound (Sphere-Packing Bound)

The number of vectors in a sphere of radius t in V_n is:

$$V(n, t) = \sum_{i=0}^t \binom{n}{i}$$

For 2^k codewords, we must have:

$$2^k \sum_{i=0}^t \binom{n}{i} \leq 2^n \implies \sum_{i=0}^t \binom{n}{i} \leq 2^{n-k}$$

Perfect Code Definition

A block code is a **Perfect Code** if it satisfies the Hamming bound with equality:

$$2^k \sum_{i=0}^t \binom{n}{i} = 2^n$$

In a perfect code, every vector in the space V_n lies in exactly one sphere of radius t around a codeword. There are no "wasted" vectors that cannot be corrected.

7.2 Examples of Perfect Codes

Only a few families of perfect codes exist:

1. **Single-Error-Correcting Hamming Codes** ($t = 1$): Satisfy: $1 + n = 2^{n-k} \implies 1 + (2^m - 1) = 2^m \quad \checkmark$
2. **The Binary Golay Code** ($[23, 12, 7]$): Corrects $t = 3$ errors.

$$\sum_{i=0}^3 \binom{23}{i} = 1 + 23 + 253 + 1771 = 2048 = 2^{11} = 2^{23-12} \quad \checkmark$$

3. The Ternary Golay Code $[[11, 6, 5]]$: Defined over $\text{GF}(3)$, corrects $t = 2$ errors.

$$\sum_{i=0}^2 \binom{11}{i} 2^i = 1 + 11(2) + 55(4) = 1 + 22 + 220 = 243 = 3^5 = 3^{11-6} \quad \checkmark$$

4. Trivial Perfect Codes: The whole space ($d_{\min} = 1$), and binary repetition codes of odd length n ($d_{\min} = n, t = (n - 1)/2$).

Quick Revision — Module 1

Linear Block Codes | PE-EC801C

Key Formulas and Metrics at a Glance

Code Rate	$R = k/n$
Hamming Distance	$d(x, y) = w(x \oplus y)$
Error Correction Limit	$t_c = \lfloor (d_{min} - 1)/2 \rfloor$
Systematic Generator Matrix	$G = [I_k \mid P]$
Parity Check Matrix	$H = [P^T \mid I_{n-k}]$
Orthogonality Condition	$G \cdot H^T = \mathbf{0} \pmod{2}$ and $c \cdot H^T = \mathbf{0}$
Syndrome Vector	$S = r \cdot H^T = e \cdot H^T$
Hamming Bound (Sphere-Packing)	$2^k \sum_{i=0}^t \binom{n}{i} \leq 2^n$
McWilliams Identity	$A^\perp(z) = \frac{1}{2^k} (1+z)^n A\left(\frac{1-z}{1+z}\right)$
Hamming Code Parameters	Length $n = 2^m - 1$, Info bits $k = 2^m - 1 - m$, $d_{min} = 3$

One-Line Definitions

- **Linear Block Code:** A code where the modulo-2 sum of any two codewords is also a valid codeword.
- **Minimum Distance (d_{min}):** The smallest Hamming distance between any two distinct codewords in a code.
- **Syndrome:** A vector calculated as $r \cdot H^T$ that indicates if errors occurred and helps locate them.
- **Coset Leader:** The vector with the smallest Hamming weight in a coset, representing the most likely error.
- **Hamming Code:** A class of perfect single-error-correcting codes with minimum distance $d_{min} = 3$.
- **Perfect Code:** A code that satisfies the Hamming (sphere-packing) bound with strict equality.
- **Dual Code ($C^\perp$):** The set of all vectors orthogonal to every codeword in the original code C .
- **Binary Symmetric Channel:** A binary input/output channel where bit-flip probability is equal for 0 and 1.

Frequently Asked / Viva Questions

Q1. What is the difference between systematic and non-systematic codes?

In systematic codes, the message bits are placed directly at the beginning or end of the codeword, making extraction trivial. In non-systematic codes, message and parity bits are interleaved or mathematically combined throughout the codeword.

Q2. Prove that $d_{min} = w_{min}$ for linear block codes.

For linear block codes, if $c_1, c_2 \in C$ are codewords, then $c_1 \oplus c_2 = c_3 \in C$ is also a codeword. The distance is $d(c_1, c_2) = w(c_1 \oplus c_2) = w(c_3)$. Thus, minimizing distance between distinct codewords is equivalent to finding the minimum weight of non-zero codewords.

Q3. What is the physical significance of the syndrome vector?

The syndrome represents the parity-check violations of the received vector. Because $S = e \cdot H^T$, it depends strictly on the channel error pattern, not on the transmitted data, allowing isolated error isolation.

Q4. Explain the ML decoding criteria over a Binary Symmetric Channel.

Maximum Likelihood decoding selects the codeword that maximizes $P(r | c)$. On a BSC with crossover probability $p < 0.5$, this probability increases as the distance $d(r, c)$ decreases. Thus, ML decoding reduces to minimum Hamming distance decoding.

Q5. Why are Hamming codes considered perfect codes?

Hamming codes have $d_{min} = 3$, meaning they correct $t = 1$ error. They satisfy the sphere-packing bound with equality since $\sum_{i=0}^1 \binom{n}{i} = 1 + n = 1 + 2^m - 1 = 2^m = 2^{n-k}$, leaving no unused vectors.

Q6. What is the role of a standard array in syndrome decoding?

A standard array groups all possible 2^n received vectors into 2^{n-k} cosets. Each coset shares the same syndrome, and the coset leader (first vector of the row) is chosen as the minimum-weight error pattern for correction.

Q7. State the McWilliams identities and explain their utility.

The McWilliams identities relate the weight enumerator polynomial of a linear code to that of its dual code: $A^\perp(z) = 2^{-k}(1+z)^n A((1-z)/(1+z))$. This allows for the simplified calculation of the weight spectrum of high-rate codes (large k) by analyzing their lower-dimension dual codes.

Q8. What is the Golay code, and why is it important?

The binary Golay code is a $[23, 12, 7]$ perfect code that corrects all patterns of up to 3 random errors. It is the only known binary perfect code that corrects multiple errors ($t > 1$) besides repetition codes.

Q9. Can a linear block code correct more errors than its minimum distance implies?

No. A code can guarantee the correction of all error patterns of weight up to $t = \lfloor (d_{min} - 1)/2 \rfloor$. While it may correct some specific patterns of weight $> t$, it cannot correct all of them.

Frequently Asked / Viva Questions (contd.)

Q10. What happens if the syndrome matches no single column in a Hamming parity check matrix?

In a Hamming code, if the syndrome is non-zero but doesn't match a column representing a single-bit error (which is impossible since columns represent all non-zero m -tuples), it indicates that more than one error has occurred, leading to decoding failure.

Comparison of Perfect and Near-Perfect Block Codes

Code Type	Parameters $[n, k, d_{min}]$	Error Correction (t)	Perfect Status
Hamming Codes	$[2^m - 1, 2^m - 1 - m, 3]$	$t = 1$	Perfect
Binary Golay Code	$[23, 12, 7]$	$t = 3$	Perfect
Ternary Golay Code	$[11, 6, 5]$	$t = 2$	Perfect
Repetition Codes	$[n, 1, n]$ (n odd)	$t = (n - 1)/2$	Perfect (Trivial)